



Big Fat Myths BABS1111 final exam notes

Big Fat Myth (University of New South Wales)



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Effect of diet on gut microbiota

Our microbiome comprises of all the microorganisms within our body. In particular, the term gut microbiome encapsulates the bacteria residing in our GIT.

The bacteria perpetually residing in our bodies is referred to as resident bacteria and has extremely beneficial properties in regard to our development, immune system and nutrition.

As these bacteria intermingle with our food, they take part in numerous vital functions. For instance, they assist with the decomposition of our food for improved absorption of nutrients. Furthermore, they help produce and discharge vital vitamins such as vitamin K and B vitamins. Example: Vitamin K is essentially exclusively formed by gut bacteria and is vital for blood clotting. Additionally, the bacteria present in our gut strengthens our immune system by fighting off foreign bacteria which could be harmful and form the basis of infections.

- Gut microbes help process food -> everyone has a different system

Majority of gas is produced by bacteria in our intestines as they digest parts of food that we cannot. Our intestine is home to trillions of bacteria -> help extract energy from our food + make vitamins + boost our immune system and play an important role of gastro intestinal barrier function, motility + development of organ systems. Gut bacteria get their nutrition primarily from undigested food such as carbohydrates and protein which come to the large intestine -> they ferment it to produce a wide range of compounds such as short chain fatty acids and gasses. Hydrogen and carbon dioxide are the most common gasses product, some people also produce methane (all odourless). Foul smell is due to volatile sulphur compounds. -> Foul smell of gas from high protein diet + increased passage of gas after high amounts of indigestible carbohydrates.

Abnormal increased flatulence -> lactose intolerance

Probiotics

Probiotics such as bacteria and yeast are microorganisms which have beneficial properties for our digestive system. Examples include dairy products and fermented foods such as yogurt, kimchi, sauerkraut and miso.

Prebiotics

Prebiotics are food which are rich in fibre (non-digestible carbohydrates) and act as food for probiotics in the gut. Stimulating selective growth of these good bacteria over bad bacteria. Examples include garlic, bananas, oats, onions and leeks.

Ways to lose weight

Eat less

A 2005 study unearthed that all the diets studied all had analogous impacts on weight loss if adhered to. Whether the diet was low-fat or low-carb, both had the outcome of a similar magnitude in weight loss. All diets tend to have their distinct advantages and disadvantages, the most optimal one is the one you enjoy and have the ability to be consistent with.

Eating less without following a diet or eating healthy

Applying 'calorie counting' you can most certainly lose weight, irrespective of the type of food consumed. The fundamental aspect to remember is to track the energy consumed and energy burned. In modern days, this is becoming increasingly important as desires for junk food increases and physical activity limits unlike our predecessors who were always on the move in search for food. However, with their being an abundance of food, and additional services like uber eats, deliveroo and menulog....these days energy in > energy out. Another factor causing people to eat more today is the release of dopamine, makes you crave food and feel rewarded.

For instance, when you compare the average energy density of 3 diets: traditional diet consumed by African subsistence farmers, average British diet (modern/western) and menu items at 3 fast food outlets. Traditional diets were less energy dense than today's means they had to eat more to achieve same energy intake but today we need less to obtain same amount of energy. However, fullness or satiety is calculated by how much food is in our stomach as hormones secrete neurons when so to signal our brains to stop eating. Therefore, you will ingest more KJs today on modern diet -> one of the reasons we overeat -> modern diets are simply more energy dense in comparison to traditional diets.

How to live a healthier lifestyle

Follow national dietary guidelines

- Make sure you enjoy a variety of nutritious foods and eat the recommended intakes of the 5 food groups daily as well as drink plenty of water.
- Plenty of vegetables (different types and colours), fruit, grain, lean meats & milk, yoghurt, cheese.

- Limit intake of foods high in saturated fat such as biscuits, cakes, cheaps and replace high fat foods with predominately polyunsaturated and monounsaturated fats such as oils, spreads and avocado.
- Limit intake of foods and drinks containing added sugars such as confectionary, soft drinks and etc.
- If you choose to drink alcohol, limit intake

Keep physically active

Cardio: Raises heart rate

- Greater effect on fat mass, fat percentage, and waist circumference when compared to resistance training alone. However, slightly decreased the lean muscle mass
- Uses more energy than resistance training and hence will raise your metabolic rate higher, you will lose more KJ -> weight.

Resistance training: Focuses on your muscles – increasing muscle strength, endurance or muscle size.

- Still did same as cardio but to lesser degree but increased the lean muscle mass
- Increases size and activity of muscle -> muscle has higher metabolic rate than adipose tissue and residual mass in the body. Therefore, increasing the size and strength of muscle within body will increase your BMR. -> burn more fat when at rest

You will 'burn' more energy doing an hour of cardio than you will doing an hour of lifting weights.

Combination of both: best results in every category.

Regular physical activity has consistently been shown to have many benefits to your health:

- Increased lifespan: by 4-5 years
- Improve well-being: reduce stress, anxiety and risk for depression
- Improve brain function & memory
- Improve quality of sleep
- Improve blood sugar and insulin levels